AQUABOTS EMI Install Procedure

Team 26061 · ENGR 498B Spring 2026 · Rev FAR-1 / WG-3 unified

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Single-document EMI install procedure for the AQUABOTS hull. Supersedes the prior `AQUABOTS_EMI_INSTALL_GUIDE.md` (Rev WG-3 / FAR-1) and `EMI_FILTERING_INSTALL_CHECKLIST.md` (FAR-1 patched). Pinouts and wire IDs match `AQUABOTS_WIRING_GUIDE.md` Rev WG-3. Where this procedure deviates from `AQUABOTS_EMI_FILTERING_BOM.pdf` Rev ISR-4b, the deviations table at the end gives the reasons.

Goal: Physically install all EMI filtering (caps, ferrites, copper tape, plate ground) on the FAR-1 / WG-3 topology. Required before extended motor testing or any auto-mode water test — motor EMF will destroy compass/GPS data otherwise.

Estimated time: 2–3 hours (soldering + per-section continuity checks + shielding). Add ~45 min for the aluminum plate shielding sub-procedure if doing it in the same session.

This document is the single source of truth for the install. Do not flip back to the predecessor docs — they have been stubbed.

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0 — Pre-Install Supply Gathering (READ FIRST)

Pull every part on this list onto the bench before unscrewing anything. Sections 0.1–0.4 are the complete shopping/staging list.

0.1 Components — caps, ferrites, beads

Quantities reflect FAR-1 reality (C5b added; F4 on P20; C4 = 25V+; F5 beads zero up-front, optional pool-test-gated). Cross-checked against BOM Rev ISR-4b — see "Known Deviations from BOM" at the end.

ID	Value / rating	Qty	Where it goes
C1	1000 µF / 35 V+ electrolytic	1	Busbar (+)/(−) studs (§1)
C2	100 nF / 50 V+ ceramic disc	~9	Busbar; ESC; ESP32 VIN; Jetson barrel; Pixhawk POWER; Starlink; servo; TDS ADC; (+1 optional 1-Wire)
C3	470 µF / 35 V+ **LOW-ESR** electrolytic	1	ESC (+)/(−) input terminals (§2)
C4	10 µF / **25 V+** electrolytic	1	ESP32 VIN / GND, <0.5" from header (§5) — **25 V+ minimum, ESP32 VIN is 12 V post-FAR-1**
C5	100 µF / 25–50 V electrolytic	1	Jetson DC barrel (§5)
C5b	47 µF / 10 V+ electrolytic	1	Pixhawk POWER DF13 +5V/GND (§6) — **NEW under FAR-1**
C6	100 µF / 16–50 V electrolytic	1	Servo 3-pin connector (§8)
C7	100 µF / 35–50 V electrolytic	1	Starlink DC plug (§7)

ID	Value / rating	Qty	Where it goes
F1	Hondark MnZn ferrite toroid 38×19×13 mm	1	P5 (8 AWG +) at ESC end, single pass (§2)
F2	Raogoodcx 13 mm clip-on ferrite	2	P9 (12V buck input) + P11 (5V buck input) (§3, §4)
F3	Raogoodcx 13 mm clip-on ferrite	1	P7 (Starlink +) at fuse end (§7)
F4	Raogoodcx 6–13 mm clip-on ferrite	1	P20 (12V buck → ESP32 pair, BOTH wires, common-mode) (§5)
F5	BOJACK ~70 Ω @ 100 MHz axial bead	**0**	**Optional / pool-test-gated.** pH is UART, not I²C — do NOT cut/splice F5 beads up-front. If pool-test reveals UART framing errors with motor on, retrofit a single CAT6 clip-on at the ESP32 end. (§9)

0.2 Wire, cable, shield, hardware

Item	Spec / size	Approx qty
Copper foil tape (conductive adhesive) — SH1	1” wide	~1 roll (10+ ft used: P20 cable + S1 cable + 4 component bodies)
Automotive self-adhesion (electrical) tape — SH2	Standard	1 roll
Split loom — SH3	1/4” and 3/8” ID	as needed
Heat shrink	Assorted (1/16“–1/2”)	full assortment
Drain wire pigtails	22–26 AWG, ~2” each	6 (one per shield)
Ring terminals	SC (M5/M6) for busbar studs	2 (for C1 leads)
Zip ties	Small	~25
Kapton tape (plate prep — §11 only)	2” wide	1 roll (on hand)
Thermal pads (plate prep — §11 only)	Dielectric silicone, ~50×50 mm sized to converter footprint (12V buck, 5V buck, BEC, ESC underside)	4–5 pads (on hand)
12 AWG plate ground jumper (§11 only)	BLK, ~6”	1

0.3 Tools

- Soldering iron (35 W+) with fine + chisel tips
- Leaded solder (60/40 or 63/37) + flux pen
- Helping-hands / PCB vise
- Flush cutters
- Wire strippers (covers 26–8 AWG)
- Crimper (ring terminals)
- Multimeter — continuity beep + DC volts ranges
- Heat gun
- Tweezers
- ESD strap
- Oscilloscope (optional but useful for V5/V6 diagnostics)
- Torque wrench for busbar studs (M6/M8 — to spec, do not over-torque ring terminals)

0.4 Consumables

- Isopropyl alcohol (90% or higher) + lint-free wipes — clean joints before soldering, clean copper tape adhesive surface
- Kapton tape — see §11 plate prep
- Dielectric grease — for the DF13 6-pin Pixhawk POWER connector pin housing only (not on contacts)
- Thermal pads — see §11 plate prep (sized per converter footprint)

1 — Prerequisites & Safety

- ☐ Main battery **DISCONNECTED** at kill switch.
- ☐ Multimeter across busbar reads **<0.1 V**. If not, find what is back-feeding before going further.
- ☐ All converters off. No LEDs lit anywhere on the plate.
- ☐ MIDI 60 A and MRBF fuse elements **removed** (re-insert during V3 smoke test).
- ☐ ESD strap on; bench has ESD-safe surface.
- ☐ Ventilation for soldering. Leaded solder fumes are not your friend.
- ☐ This procedure document open. Predecessor `AQUABOTS_EMI_INSTALL_GUIDE.md` and `EMI_FILTERING_INSTALL_CHECKLIST.m` have been stubbed and should not be referenced.
- ☐ `AQUABOTS_WIRING_GUIDE.md` Rev WG-3 nearby for layout reference (this procedure embeds wire IDs but the wiring guide governs routing).
- ☐ BOM PDF `AQUABOTS_EMI_FILTERING_BOM.pdf` Rev ISR-4b nearby — but treat as stale. See "Known Deviations from BOM" at end of this document.

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2 — Reference: Wire IDs and Pinouts (embedded so you don't flip)

FAR-1 / WG-3 power topology (four ATO branches, no dedicated 5V Power Module).

2.1 Wire IDs touched in this procedure

Wire ID	Path	Gauge	Rail / signal	Used in section
P5	(+) Busbar → MIDI 60 A → ESC (+) input	8 AWG RED	22–26 V (busbar)	§2 (ESC branch)
P6	(−) Busbar → ESC (−) input	8 AWG BLK	GND	§2
P7	(+) Busbar → ATO 7.5 A → Starlink DC (+)	12 AWG RED	22–26 V	§7 (Starlink)
P8	(−) Busbar → Starlink DC (−)	12 AWG BLK	GND	§7
P9	(+) Busbar → ATO 3 A → 12V buck input (+)	12 AWG RED	22–26 V → 12V buck	§3 (12V buck input)
P10	(−) Busbar → 12V buck input (−)	12 AWG BLK	GND	§3
P11	(+) Busbar → ATO 2 A → 5V buck input (+)	12 AWG RED	22–26 V → 5V buck	§4 (5V buck input)
P12	(−) Busbar → 5V buck input (−)	12 AWG BLK	GND	§4
P13	(+) Busbar → ATO 2 A → BEC input (+)	12 AWG RED	22–26 V → BEC	§8 (BEC/servo)
P14	(−) Busbar → BEC input (−)	12 AWG BLK	GND	§8
P17	ESC 3-phase → Motor 5080	8 AWG ×3	3-phase motor	(do not wrap; noise source)
P18	12V buck → Jetson DC barrel	22 AWG pr	12 V	§5
P19	12V buck → Cooling fan	22 AWG pr	12 V	(no extra filter)
P20	12V buck → ESP32-S3 breakout VIN	22 AWG pr	**12 V** (FAR-1)	§5 (F4 + C4 + C2); §10 (copper-tape wrap)
P21	5V buck → Pixhawk 4X **POWER** (DF13 6-pin)	22 AWG pr	5.0 V	§6 (C5b + C2)
P22	BEC (Castle) → Rudder Servo	22 AWG pr	5.5 / 6 V (BEC spec)	§8

Wire ID	Path	Gauge	Rail / signal	Used in section
S1	ESP32 UART1 → Pixhawk (MAVLink)	22 AWG	3.3 V UART	§10 (copper-tape wrap)

2.2 Pixhawk POWER (DF13 6-pin) pinout

Standard ArduPilot/PX4 power port. C5b + C2 install across pins 1 and 6 (or on the harness pair within 1" of the connector).

Pin	Signal	Notes
1	VCC (+5 V)	C5b (+) and C2 land here
2	VCC (+5 V) (same rail)	
3	V_BAT analog	**Floating** (no Power Module under FAR-1)
4	I_BAT analog	**Floating** (no Power Module under FAR-1)
5	GND	
6	GND	C5b (−) and C2 land here

Pixhawk telemetry V/I will read 0 — accepted under FAR-1 mission profile per WG-3 §2.4.

2.3 ESP32-S3 breakout — VIN / GND

C4 + C2 land directly across the **VIN** and adjacent **GND** header pins on the Meshnology breakout. Within 0.5" of the pins.

VIN is 12 V post-FAR-1, not 5 V. This is the source of the C4 = 25 V+ requirement.

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3 — Universal Rules (read once, apply everywhere)

1. **Parallel caps share the same solder points.** Every electrolytic in this procedure is paired with a 100 nF ceramic on the same junction. Both must be present — the electrolytic catches low-frequency, the ceramic catches HF.
2. **Electrolytic polarity.** Long lead / no stripe → (+). Stripe / shorter lead → (−). Reversed = popped cap, sometimes violently. Check every one before energizing.
3. **Ceramics have no polarity.**
4. **Voltage rating is a ceiling, not a target.** 50 V cap on a 5 V rail is fine. Busbar-side caps need 50 V+ minimum to survive ESC transients (rail is 22–26 V plus spikes).
5. **Clip-on ferrites = (+) wire only.** Single exception: **F4** on P20 takes both (+) and (−) together for common-mode filtering.
6. **F1 toroid = single pass.** One wire through the hole, no multi-turn at these currents.
7. **F5 axial beads = inline.** Cut the signal wire, solder bead in series, heat-shrink. **Not used up-front under FAR-1** — pH is UART, not I²C. Pool-test-gated only.
8. **Lead length is everything.** Every mm between a cap and the pin it protects is series inductance that kills HF filtering. Cut leads short.
9. **Copper tape shields grounded at ONE end only** (source). Ungrounded = antenna. Both ends = ground loop.
10. **Install order:** busbar caps → input ferrites → wire the branches → output ferrites → output caps → sensor filtering → copper tape last → multimeter test → reinsert fuses (MIDI then MRBF last).
11. **Per-section continuity check.** After every numbered section below, multimeter (+) → (−) at the install point. Must read **open (>1 MΩ)** before moving on. Every section ends with this gate.
12. **Secure everything.** Marine vehicle = vibration. Zip-tie or hot-glue every cap and ferrite so nothing swings.

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4 — Step-by-Step Procedure

§1 — Busbar Bypass (DO THIS FIRST)

Highest-priority filter. Catches ESC switching at the source bus.

- ☐ **1.1** Verify busbar dead: multimeter across (+)/(−) studs reads **<0.1 V**.
- ☐ **1.2** Prep **C1 (1000 µF / 35 V+ electrolytic):** crimp ring terminals to leads, leads <1".
- ☐ **1.3** Mount C1 across (+) and (−) busbar studs. **Polarity:** long lead / no stripe → (+) stud.
- ☐ **1.4** Prep **C2 (100 nF / 50 V+ ceramic):** leads trimmed to <0.5".
- ☐ **1.5** Mount C2 in PARALLEL with C1 — same two studs, same solder joints. No polarity.
- ☐ **1.6** Torque busbar studs to spec (do not over-torque). Zip-tie both caps so they cannot swing under vibration.
- ☐ **1.7 Continuity gate:** (+) → (−) busbar must read **open (>1 MΩ)**. If it shorts, stop and inspect — almost always a reversed C1 or a stray strand.

  Common mistakes: C1 reversed (long lead landed on (−)). C2 missing — busbar then filters low-freq only. Lead-length too long — under-torqued ring terminal forced you to leave 1.5" of bare lead, which becomes an inductor.

§2 — ESC / Motor Branch (F1 + C3 + C2)

Single biggest noise source in the system. Do not skip any of it.

- ☐ **2.1** Identify the **8 AWG RED (+)** lead **P5** between MIDI 60 A fuse and ESC input.
- ☐ **2.2** Disconnect P5 at ESC end. Pass the 8 AWG wire through **F1 (Hondark 38×19×13 mm MnZn toroid) — single pass, one time through.** Do NOT wrap.
- ☐ **2.3** Position F1 within **2"** of ESC power input. Zip-tie F1 to the 8 AWG or adjacent structure so it cannot slide.
- ☐ **2.4** Reconnect 8 AWG to ESC (+) terminal.
- ☐ **2.5** Prep **C3 (470 µF / 35 V+ LOW-ESR electrolytic** — the **only** cap where low-ESR matters). Leads <0.5".
- ☐ **2.6** Solder C3 directly across ESC (+) and (−) power input terminals. **Polarity matters.**
- ☐ **2.7** Solder **C2 (100 nF / 50 V+ ceramic)** in PARALLEL with C3 — same terminals.
- ☐ **2.8** Heat-shrink the joints. Zip-tie.
- ☐ **2.9 Continuity gate:** ESC (+) → (−) must read **open**, not shorted.

  Common mistakes: Multi-turn through F1 (don't — single pass at this current). Substituting standard-ESR for C3 (the high-current transients need low-ESR specifically here). F1 too far from ESC (must be <2"; the wire between the toroid and the ESC input is unfiltered).

§3 — 12V Buck INPUT (F2 on P9)

12V buck now feeds three loads: Jetson DC barrel, cooling fan, and ESP32-S3 VIN. Input-side filter happens here; output-side filtering is split between the loads (§5).

- ☐ **3.1** Identify **P9** — 12 AWG RED (+) wire from ATO 3 A SB fuse to 12V buck input.
- ☐ **3.2** Snap **F2 (Raogoodcx 13 mm clip-on)** around the **(+) wire only** (not both wires). Within **2"** of the buck input.
- ☐ **3.3** Close clip-on, secure with tape or zip tie.
- ☐ **3.4 Continuity gate:** P9 (+) at buck input → P10 (−) reads **open**.

  Common mistakes: Snapping the clip-on around both P9 and P10 — that's common-mode filtering, which is wrong here. The ESC-side noise is differential and you need it suppressed differentially. Common-mode is reserved for the ESP32 output (F4, §5).

§4 — 5V Buck INPUT (F2 on P11)

Mirror of §3 for the Pixhawk-feeding rail.

- ☐ **4.1** Identify **P11** — 12 AWG RED (+) wire from ATO 2 A SB fuse to 5V buck input.
- ☐ **4.2** Snap **F2 (second 13 mm clip-on)** around the **(+) wire only**. Within **2"** of the 5V buck input.
- ☐ **4.3** Close, secure.
- ☐ **4.4 Continuity gate:** P11 (+) → P12 (−) at buck input reads **open**.

§5 — 12V Buck OUTPUT (F4 on P20, C4 + C2 at ESP32, C5 + C2 at Jetson)

The 12V buck output fans three ways. The ESP32 leg gets its own common-mode ferrite plus output caps; the Jetson barrel gets output caps; the fan needs nothing.

5a. Common-mode ferrite F4 on the ESP32 leg

- ☐ **5.1** Identify **P20** — 22 AWG pair from 12V buck output to ESP32-S3 VIN.
- ☐ **5.2** Snap **F4 (Raogoodcx 6–13 mm clip-on)** around **BOTH (+) and (−)** wires of P20 **together**. Common-mode filter. Within **2"** of the ESP32 board.
- ☐ **5.3** Close, secure. (Sole exception to the "+ only" rule — see Section 3 / Universal Rules, rule 5.)

5b. Output caps at ESP32 VIN

- ☐ **5.4** Prep **C4 (10 µF / 25 V+ electrolytic)** + **C2 (100 nF ceramic)**. Leads as short as possible. > **C4 voltage rating gate:** ESP32 VIN is 12 V post-FAR-1 (not 5 V as the BOM says). A 10 V-rated cap will fail. Verify the cap on your bench reads **25 V** or higher before soldering.
- ☐ **5.5** Solder C4 + C2 in PARALLEL across the ESP32 **VIN** and **GND** header pins. C4 polarity matters. Within **0.5"** of the header pins.
- ☐ **5.6 Continuity gate:** ESP32 VIN → GND reads **open**.

5c. Output caps at Jetson DC barrel

- ☐ **5.7** Prep **C5 (100 µF / 25–50 V electrolytic)** + **C2 (100 nF ceramic)**.
- ☐ **5.8** Solder C5 + C2 in PARALLEL across the Jetson 12 V DC barrel input pigtail (before the barrel plug). C5 polarity matters. Within **1"** of the barrel.
- ☐ **5.9 Continuity gate:** Jetson 12 V input → GND reads **open**.

Fan runs off the same 12V output via its JST connector — no extra filtering by design.

Common mistakes: C4 = 10 V cap from BOM stock — installs and dies on first power-on (see Deviations table). F4 around only one wire — that's a differential-mode filter, but the ESP32 leg's worst noise is common-mode pickup from radiated ESC EMF. Solder joints >0.5" from ESP32 header pins — leads become inductors that kill HF filtering at the radio.

§6 — 5V Buck → Pixhawk POWER Branch (C5b + C2 at DF13)

FAR-1 NEW. The dedicated 5V Power Module is gone. The generic 5V buck has no internal filtering — this branch needs its own bulk + HF bypass at the DF13 connector or you will see Pixhawk brownouts on ESC arm.

- ☐ **6.1** Locate the Pixhawk **POWER** port (DF13 6-pin).
- ☐ **6.2** Prep **C5b (47 µF / 10 V+ electrolytic)** + **C2 (100 nF ceramic)**. Leads as short as possible.
- ☐ **6.3** Solder C5b + C2 in PARALLEL across **+5 V (pin 1)** and **GND (pin 6)** of the DF13 6-pin connector — or on the P21 harness pair within **1"** of the connector. C5b polarity matters.
- ☐ **6.4 Continuity gate:** +5 V → GND at the connector reads **open**, not shorted.

Brownout watch (post-install): If Pixhawk resets during ESC arm or motor calibration, the generic buck is sagging under inrush. Mitigations in priority order: (1) Add a Schottky series diode + larger bulk cap at the POWER input; (2) revert this branch to a dedicated Pixhawk Power Module (and remove C5b + C2 — the Power Module reintroduces internal filtering).

Common mistakes: Skipping C5b ("the buck regulates fine on the bench"). Yes — until ESC inrush slams the rail. The bulk cap is what catches the sag.

§7 — Starlink Branch (F3 + C7 + C2)

- ☐ **7.1** Identify **P7** — 12 AWG RED (+) from ATO 7.5 A SB fuse to Starlink DC input.
- ☐ **7.2** Snap **F3 (13 mm clip-on)** around (+) wire only. Within **3"** of the **fuse block** (catch the noise near the source — the Starlink run is long).
- ☐ **7.3** Prep **C7 (100 µF / 35–50 V electrolytic)** + **C2 (100 nF ceramic)**.
- ☐ **7.4** Solder C7 + C2 in PARALLEL across the Starlink DC connector pigtail. C7 polarity matters. Within **1"** of the connector. Heat-shrink the joints.

☐ **7.5 Continuity gate:** Starlink (+) → (−) at the plug reads **open**.

Common mistakes: F3 placed at the Starlink end instead of the fuse end — Starlink has its own internal conditioning, so the goal is to prevent ESC noise from coupling onto the long P7 run, which means catching it at the source.

§8 — BEC → Servo Branch (C6 + C2, no ferrite by design)

☐ **8.1** Prep **C6 (100 µF / 16–50 V electrolytic) + C2 (100 nF ceramic)**.
☐ **8.2** Solder C6 + C2 in PARALLEL across the (+) and (−) wires of the **P22** servo cable. **At the servo end**, not the BEC end (the servo is the inductive noise source; the cap absorbs there). C6 polarity matters. Within **1"** of the servo 3-pin connector.
☐ **8.3** Heat-shrink the junction.
☐ **8.4 Continuity gate:** servo (+) → (−) reads **open**.

No input ferrite on this branch by design — see WG-3 §4.2.

§9 — Sensor Signal Filtering (TDS critical, pH gated, 1-Wire optional)

FAR-1 reality check: pH path is **UART @ 9600 baud** per `firmware/main/sensor_bridge.c` — not I²C. F5 BOJACK beads are **not installed up-front**. Do not cut and splice beads inline. If pool-test reveals UART framing errors with motor running, retrofit a single 13 mm clip-on on the CAT6 at the ESP32 end. Document and re-verify.

9a. CRITICAL — TDS analog (S10)

☐ **9.1** Solder **C2 (100 nF ceramic)** between the ESP32 ADC pin reading TDS (S10) and ESP32 GND. **Within 0.25" of the ESP32 pin** — this is the strictest distance gate in the procedure. ADC accuracy depends on it.
☐ **9.2** Solder direct to header pin or breakout pad. Shortest possible leads.
☐ **9.3 Continuity gate:** TDS pin → GND reads **open** (cap acts as open at DC).

9b. SKIP up-front — pH UART (S8)

☐ **9.4 DO NOT install F5 beads on pH lines.** Do not cut/splice the CAT6 pH pair. This step is intentionally a no-op at install time. Reassess after pool test (§troubleshooting).

9c. OPTIONAL — 1-Wire DS18B20 (S9)

☐ **9.5** *(Optional, defer unless visible noise.)* Solder C2 (100 nF ceramic) between the ESP32 1-Wire data pin (S9) and GND, **<0.5" from the pin**.

9d. DO NOT INSTALL — S7 IMU, S11 GPS

☐ **9.6** S7 (IMU) and S11 (GPS) are removed in WG-2 (replaced by Pixhawk built-in IMU and M8N GPS). Any beads or caps the BOM specified for these lines are **not needed** — skip entirely.

Common mistakes: TDS cap landed on a long jumper to GND — that 1" of wire kills the filter. C2 must reach the ADC pin and the nearest GND pin in well under a quarter-inch.

§10 — Copper Tape Shielding (DO LAST — after all filters installed and bench-verified)

Copper tape goes innermost (the conductive layer) → automotive self-adhesion tape (insulation, prevents shorts to neighbors) → split loom (mechanical protection). Drain wire grounded at **ONE end only** (source). Leave **0.5"** bare at connectors so the tape doesn't short to pins.

10a. Cable-body wraps

☐ **10.1 P20** (12V buck → ESP32, 22 AWG pair):
- Wrap the cable body in copper tape. Slight overlap, no gaps.
- Solder a **2"** drain pigtail (22–26 AWG) to the copper tape **at the 12V buck (source) end**.
- Land the drain pigtail on the **12V buck GND** terminal.
- Leave the ESP32 end of the tape **floating** (no drain).

  - Cover the entire copper tape run with automotive self-adhesion tape (SH2) so it can't short to neighbors.
- ☐ **10.2 S1** (ESP32 UART → Pixhawk JST-SH 4-pin):
  - Same procedure as P20 but smaller cable.
  - Drain at the **ESP32 (source) end → ESP32 GND** at the breakout.
  - Pixhawk end floating.
- ☐ **10.3 CAT6 sensor bundle** (S8/S9/S10): foil shield is built in. Verify the CAT6 foil drain is landed at **ESP32 GND only** (bow end floating). Should already be correct per WG-3 §3.2 — verify, do not re-route.

10b. Component-body wraps (NEW under WG-2 — all to (−) busbar, ONE end only) For each component below, wrap the full body of the device in copper tape, then automotive self-adhesion tape over it. Solder a **1–2"** drain pigtail (22–26 AWG) from the copper tape surface to the **(−) busbar**. **One drain per body. Never daisy-chain drains.**

- ☐ **10.4 ESC body** — replaces the original heat shrink. Avoid covering heatsink fins (don't thermally trap the ESC). Drain → (−) busbar.
- ☐ **10.5 BEC body** (Castle). Drain → (−) busbar.
- ☐ **10.6 12V buck converter body.** Drain → (−) busbar.
- ☐ **10.7 5V buck converter body.** Drain → (−) busbar.

10c. After every copper-tape install

- ☐ **10.8** Cover all copper tape with automotive self-adhesion tape (SH2). Bare copper tape near other wiring will short something at the worst possible time.
- ☐ **10.9 Continuity check per shield:** drain wire → its destination GND = **closed** (through ground path). Drain wire → any (+) rail = **open**. If a drain reads continuity to (+), the tape is touching a (+) terminal somewhere — find and insulate it.

10d. Do NOT wrap

- CAT6 bow sensor bundle (S8/S9/S10) — foil shield handles it.
- P5/P6 ESC-motor 8 AWG — noise source, not victim. F1 + C3 + C2 handle it.
- P17 3-phase ESC → motor wires — noise source. Split loom only.
- P1–P4 4 AWG trunk — short, fat, low impedance; not worth shielding.
- S7, S11 — sensors removed in WG-2.
- S10 TDS individually — already inside the CAT6 foil.

Common mistakes: Drain at both ends → ground loop, makes things worse. No insulation over copper tape → drain shorts to a neighbor terminal. Daisy-chained drains → multiple component-body shields share one inductive return path, defeating the point.

§11 — Aluminum Plate Shielding (referenced from WG-3 §5; status: design only, parts on hand)

The grounded 12×24" aluminum mounting plate is a conductive shield between ESC (mounted underneath) and topside electronics (Jetson, Pixhawk, ESP32, converters). **Status per WG-3 §5: DESIGN ONLY, parts on hand, not yet implemented.** Bundled here because it's directly part of the EMI work; if you're skipping this section in the same install pass, mark §11 deferred and proceed to §12.

11a. Plate prep (before mounting topside components)

- ☐ **11.1** Apply **Kapton tape** to the entire **top surface** of the aluminum plate. Double-layer wherever a ring terminal or bolt head will sit.
- ☐ **11.2** Install **dielectric thermal pads** under each converter footprint (12V buck, 5V buck, BEC).
- ☐ **11.3** Prep a **12 AWG BLK plate-ground jumper**: ring terminal → plate bolt at one end, the other end prepped for the (−) busbar.
- ☐ **11.4** Land the plate ground jumper on the **(−) busbar**. **Single point only — no second plate-to-ground anywhere on the boat.** A second tie = ground loop, kills the filtering scheme.

11b. ESC under-plate mounting

- ☐ **11.5** Apply thermal pad to ESC underside. Mount ESC to the plate's **bottom surface** at the stern-ward end.

☐ **11.6** Drill/grommet a hole for the 8 AWG (P5/P6) power feed. Route through the grommet — wires must not contact bare plate edges.
☐ **11.7** Drill a separate small grommet hole for **S3 PWM** (Pixhawk → ESC). Route around plate edge or through this dedicated hole — keep separate from the 8 AWG power feed.
☐ **11.8** Conformal-coat the ESC PCB **OR** wrap the ESC body in closed-cell foam splash guard. ESC is now the lowest component in the hull and **will** see splash. Waterproofing is mandatory, not optional.

Critical rules (re-stated for emphasis): - **Single ground point only** — no second plate-to-ground. - Kapton goes **under** every topside component. The plate is now live ground; a scraped trace or bare ring terminal edge = dead short. - ESC waterproofing is mandatory.

§12 — Final Routing & Separation

☐ **12.1** Verify signal wires (22–26 AWG, CAT6) are physically routed **at least 12"** from 8 AWG motor power. Use opposite hull walls (signals starboard, power port — per WG-3 §6).
☐ **12.2** Confirm S3 PWM goes through its own grommet hole, not the 8 AWG hole.
☐ **12.3** Verify no copper tape edges contact unintended surfaces. Add SH2 patches anywhere a tape edge is close to a (+) terminal.
☐ **12.4** Tug-test every soldered joint. Re-zip-tie any cap that can swing.

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5 — Post-Install Verification (V1–V6)

Do all six in order. Stop on the first failure; do not skip ahead.

☐ **V1. Visual inspection.** No loose solder, no pinched wires, all caps secured, no copper tape touching unintended surfaces. Polarity of every electrolytic visually re-checked.
☐ **V2. Full busbar continuity.** (+) → (−) on the busbar = **open (>1 MΩ)**. If shorted anywhere, stop and find it before re-energizing.
☐ **V3. Smoke test.** Reconnect battery at kill switch (insert MIDI 60 A, then MRBF **last**). Look for smoke. (Yes, really.)
☐ **V4. Per-rail voltage.** Multimeter at each branch output:
  - Busbar = **22–26 V** (depending on demo vs production battery state).
  - 5V buck output to Pixhawk POWER (P21) = **5.0 ± 0.1 V**.
  - 12V buck output to Jetson + ESP32 + fan = **12.0 ± 0.2 V**.
  - BEC output (P22) = per BEC spec (typically 5.5 V or 6 V).
  Removed under FAR-1: the old "Pixhawk 5V rail from Power Module" check no longer applies. Pixhawk POWER is now fed by the 5V buck (P21) and is checked above.
☐ **V5. Motor spin-up.** Power on motor briefly with **prop removed**, low throttle. No smoke, no abnormal buzz, no rail sag >0.5 V on any branch.
☐ **V6. Compass-drift gate.** Open QGC, watch heading. With motor OFF: heading is stable. With motor running at low throttle: heading shifts **<2°** from static value. **>5° shift = EMI is still a problem** — re-inspect §2 (ESC branch) and §11 plate ground.

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6 — Update Docs After Install (deferred — track but not in scope today)

- Update `AQUABOTS_EMI_FILTERING_BOM.pdf` status column (ORDER → INSTALLED). Deferred — BOM Rev ISR-5 regen is a separate task.
- Update `AQUABOTS_POWER_SYSTEM_CHANGELOG.md` with install date and append Rev ISR-5 block.
- Bump `AQUABOTS_WIRING_GUIDE.md` WG-3 → WG-4 with summary: "EMI filtering installed. Production DSServo + new BEC wired. Compass + accel recalibrated post-install."
- Record SERVO1 calibration values (MIN/MAX/TRIM/REVERSED) in the wiring guide or a new `PIXHAWK_SERVO_CONFIG.md`.

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7 — Known Deviations from BOM Rev ISR-4b

BOM Rev ISR-4b says	This procedure does	Reason
4 ferrite beads on I²C (2 pH SDA + 2 SCL) + 2 optional UART	**0 F5 beads up-front.** Pool-test-gated CAT6 clip-on only.	Per `firmware/main/ser` pH is UART @ 9600 baud. I²C beads not needed.
2 ferrite beads on IMU SDA/SCL	**0** — sensor removed.	S7 IMU removed in WG-2. Pixhawk built-in IMU.

BOM Rev ISR-4b says	This procedure does	Reason
Copper tape on P21, S10 individual, optional S1/S2	**P20** (not P21) + S1 + ESC body + BEC body + 12V buck body + 5V buck body.	FAR-1 moved ESP32 to 12V buck rail (P21 → P20). WG-2 added component-body wraps for radiated emission. CAT6 foil already shields S10, so individual wrap removed.
5V Power Module on Pixhawk POWER (internal filtering)	**Generic 5V buck + C5b 47 µF/10 V+ + C2 100 nF.**	FAR-1 consolidation — Power Module removed, generic buck has no internal filtering.

BOM Rev ISR-4b says	This procedure does	Reason
C4 = 10 µF / 10 V+ on ESP32 VIN	**C4 = 10 µF / 25 V+** electrolytic.	ESP32 VIN is 12 V post-FAR-1, not 5 V. 10 V cap would fail.
F4 location: "5V Buck output to ESP32"	**F4 on P20 (12V Buck output to ESP32 pair).**	ESP32 moved from 5V buck rail to 12V buck rail under FAR-1.
C2 ceramic qty "10+"	~9 used (~10–12 with optional 1-Wire and spares). BOM qty is sufficient.	Qty unchanged in spirit; reflect actual install.

These deviations feed into BOM Rev ISR-5 regeneration (separate task, not in scope for this install).

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8 — Troubleshooting & Fallback

If V1–V6 fail, follow this triage tree before reaching for a different design.

Symptom	First place to check	Mitigation
V2 fails — busbar shorted	C1 polarity reversed, or stray wire strand across studs.	Disconnect C1, re-test. If now open, C1 was reversed; replace.
V3 fails — smoke / smell on power-up	Reversed electrolytic somewhere; or copper tape shorting.	Kill power. Identify which converter is hot. Inspect every electrolytic in that branch. Re-check copper-tape insulation per §10c.
V4 fails — Pixhawk 5 V rail low (<4.9 V)	C5b missing or sagging buck under load.	Re-verify §6. If C5b is in place: add Schottky series + larger bulk cap, or revert to dedicated Pixhawk Power Module.

Symptom	First place to check	Mitigation
V4 fails — ESP32 12 V rail low	F2 wired around both wires (= common-mode, wrong here).	Re-do §3 with F2 around (+) only.
V5 fails — Pixhawk reset on ESC arm/calibration	Brownout: 5V buck sagging during inrush.	Per §6 brownout watch — Schottky + bulk, or revert to Power Module.
V5 fails — abnormal motor buzz	F1 not single-pass (multi-turn = saturated core).	Re-do F1 as single pass.
V6 fails — compass drift >5° with motor on	ESC body not shielded; F1 too far from ESC; copper-tape drain bonded at wrong end.	Re-verify §2.3 (F1 within 2”), §10.4 (ESC body wrapped, drain to (−) busbar at one end), §11.4 (plate ground = single point).
Pool test — pH UART framing errors with motor on	Conducted noise on CAT6.	Add **one** 13 mm clip-on ferrite on the CAT6 bundle at the ESP32 end. Do **not** cut/splice F5 beads inline. Document and re-run V6.
Pool test — TDS reading noisy with motor on	C2 at ADC pin too far from pin, or missing.	Re-solder C2 to <0.25” from the ADC pin. If still noisy after that, add a clip-on ferrite on the CAT6 at the ESP32 end (same retrofit as pH).
Compass drift fixed by detaching plate ground	Ground loop — second plate-to-ground tie somewhere.	Find the second tie. Per §11.4 only **one** plate-to-(−)-busbar connection.

Optional pool-test-gated additions (only if symptoms appear)

- CAT6 clip-on ferrite at ESP32 end — for pH UART or TDS noise (one ferrite covers both since they share the bundle).
- Schottky + larger bulk cap at Pixhawk POWER input — for buck brownout under ESC inrush.
- Revert 5V buck → dedicated Pixhawk Power Module — last resort, restores telemetry V/I and internal filtering.

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9 — Do-Not-Do List (re-stated)

- Do **NOT** energize anything until **all** §-end continuity gates pass.
- Do **NOT** ground both ends of any shield. Source end only.
- Do **NOT** multi-turn the F1 toroid. Single pass only.
- Do **NOT** swap F4 from common-mode (both wires) to single-wire. Topology is intentional.
- Do **NOT** skip the 100 nF ceramic in any parallel pair. Electrolytic alone covers low-frequency only.
- Do **NOT** cut and splice F5 beads inline on pH SDA/SCL — pH is UART, not I²C.
- Do **NOT** use a 10 V-rated cap on the ESP32 VIN rail — it’s 12 V post-FAR-1.
- Do **NOT** skip C5b at the Pixhawk POWER connector — the old Power Module’s internal filtering is gone.
- Do **NOT** add a second plate-to-ground tie — single point only or the whole filtering scheme falls apart.

Companion documents (still authoritative for their domains):

- `AQUABOTS_WIRING_GUIDE.md` Rev WG-3 — wire IDs, connector pinouts, rail layout, routing.
- `AQUABOTS_EMI_FILTERING_BOM.pdf` Rev ISR-4b — parts catalog (with deviations per §7 above).
- `AQUABOTS_POWER_SYSTEM_CHANGELOG.md` — full revision history.

Predecessor docs (stubbed, do not reference):

- `AQUABOTS_EMI_INSTALL_GUIDE.md` Rev WG-3 / FAR-1 — merged into this procedure.
- `field-procedures/EMI_FILTERING_INSTALL_CHECKLIST.md` FAR-1 patched — merged into this procedure.